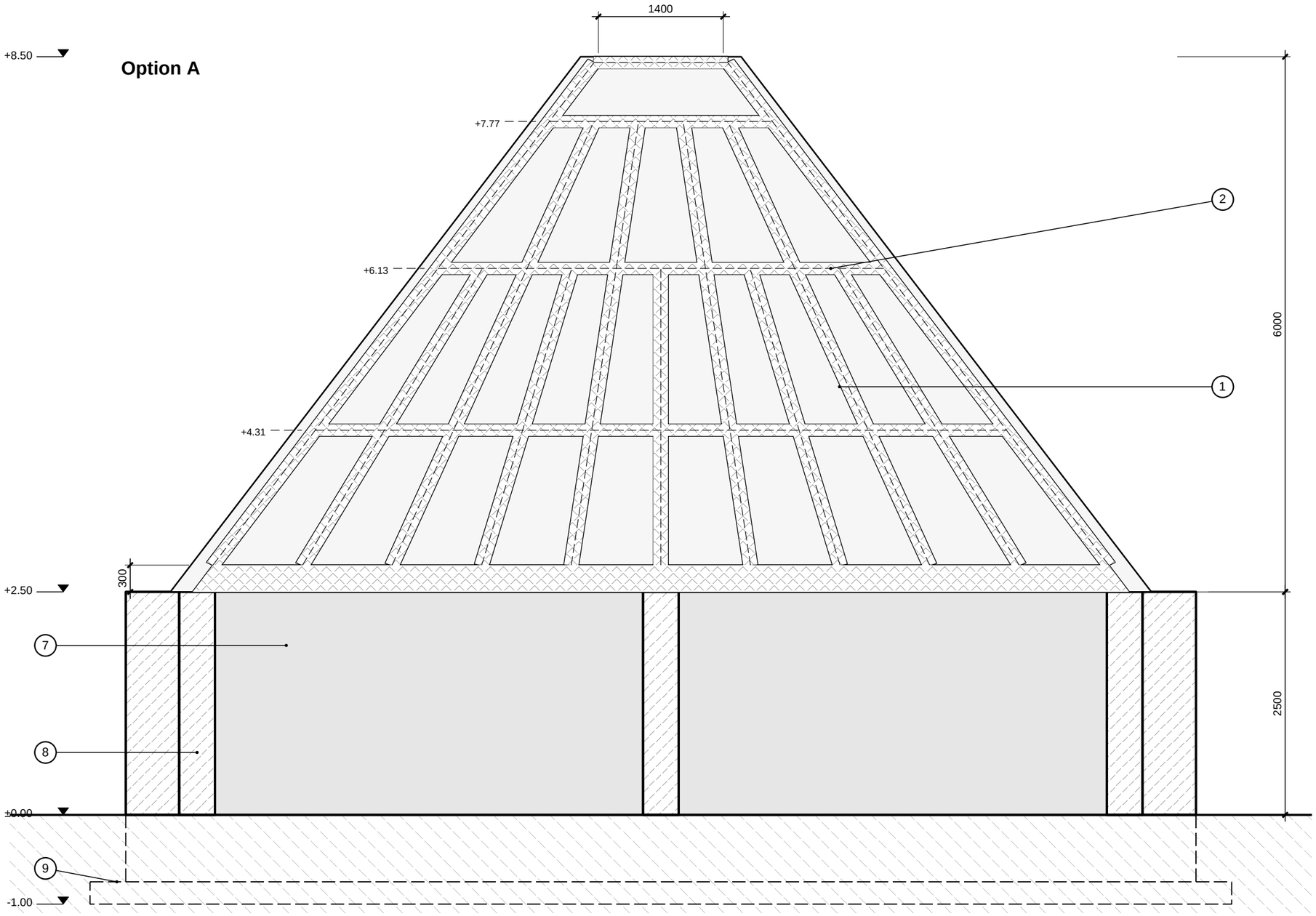
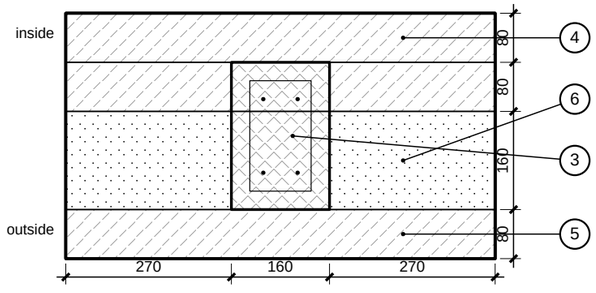


Section through one rib, square to the slope, 1:10



Roof grid A: ruled-surface columns

one roof face seen square on

(1) Columns run in straight lines from the base edge (every 1050) towards the crown edge, so they converge like the ruled surface of the sketch.

(2) When the spacing drops to 500 (about +6.13) every other column stops; at about +7.77 only the four hip columns go on to the crown ring.

(3*) Rib (column): printed channel 160 wide x 240 deep in the 160 air zone + layer 2, filled with cast concrete, 4 Ø16 + Ø8 stirrups every 200 (assumed).

(4) Inner layer 1, 80: continuous, covers the grid on the room side.

(5) Outer skin 80: continuous, covers the grid on the outside (cladding on it).

(6*) Between the ribs the air zone stays open for thermal insulation (dotted).

(7) Printed wall 400 (same layers as the roof), to +2.50.

(8) Buttresses 400 wide, 1000 from the inner face (600 outside), at the corners and mid-span; roof ribs and wall bars tie into the solid base above them.

(9*) Strip footing 1800 x 250, underside at -1.00 (hidden, dashed).

* = assumed, to confirm

Notes

Columns stop on a horizontal rib (at 3626 and 5275 above the eave); bars are lapped / hooked into it.

Horizontal ribs at 1813, 3626, 5275, plus the 300 solid base (bottom ring) and the top ring at the crown.

Rough quantities, four faces (hips counted twice, so slightly high): ribs 325 m, 4 Ø16 each bars, steel 2049 kg, cast concrete 12.5 m3.

Rib size, bar count and the 160 x 240 channel are assumptions for the engineer to check. Elevation shows the face square on; true lengths are 1.25 x the drawn heights.

0 1 2 3 4 5 m

Scale 1:50 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered